

Diabetes Technology Expert Program

Insulin Pumps: the Master Module

1. Introduction

Welcome to this journey into the world of insulin pumps!

Insulin pumps are revolutionizing diabetes management, allowing individuals to fine-tune their insulin dosages, mimicking the operations of a healthy pancreas. Compared to multiple daily injections, insulin pump therapy is linked to reductions in HbA1c levels, fewer episodes of hypoglycemia, tighter glucose control, and typically, an enhancement in the quality of life as insulin can be dosed more easily.

Modern medical technology has brought us automated insulin delivery systems, linking your insulin pump, a glucose sensor, and an algorithm. This marvel of innovation adjusts your insulin delivery every five minutes based on sensor-detected glucose levels. These automated insulin delivery systems are so effective, that they are recommended for all people with type 1 diabetes and those with insulin-deficient diabetes.

However, transitioning to an insulin pump isn't always straightforward. A thorough understanding of how the pump operates, its usage, and potential risks is essential. Ensuring accurate and safe insulin dosing demands precise pump settings and prompt action in case of malfunctions or blockages.

This comprehensive module on insulin pumps is here to guide you. The videos are created by Dr. Inge Van Boxelaer, Diabetotech's founder and an endocrinologist at the St-Lucas Hospital in Ghent in Belgium. All content has been reviewed by Dr. Liesbeth Van Huffel, endocrinologist at the OLV Hospital in Aalst in Belgium, and Melissa Holloway from the UK and Renate Groenendijk from the Netherlands, two individuals living with type 1 diabetes.

Be assured, all the information provided here is completely independent from the insulin pump manufacturing companies, ensuring a genuine perspective on the advantages and drawbacks of different insulin pumps.

Before we delve into the details, it is important to note that these videos are not intended as medical advice. Always consult with your healthcare provider before making any adjustments to your diabetes therapy.

In the next 45 minutes, you will learn:

- The available insulin pumps and their key differences,
- How an insulin pump works and how to set basal rates and bolus insulin ratios,
- What to keep in mind when changing the insulin and infusion set,
- How to respond to an alarm,
- And how people with diabetes experience the transition from pens to a pump.

This module will equip you with insights into the typical start-up education of an insulin pump, enabling you to have more confidence to initiate or guide others with an insulin pump.

Enjoy the journey!

2. Comparison of different insulin pumps

Welcome to this lesson, where we will explore the various insulin pumps available in the market.

In this video, we will cover tubed pumps and patch pumps, and essential considerations to keep in mind while choosing the right insulin pump for your needs.

Insulin pumps, small yet power-packed devices, carry within them a reservoir of insulin, a pumping mechanism, a tiny computer and a battery. They work with rapid-acting insulin, delivering it at a steady, low rate as basal insulin, or in larger doses as a bolus during meals or when a correction dose is needed. While the insulin used is generally Novorapid, depending on the manufacturer's labelling and/or a healthcare professional's discretion, Humalog, NovoRapid, NovoLog, Apidra, Fiasp, or Lyumjev may also be used.

Let's delve into the intriguing differences among various insulin pumps.

Essentially, we have two categories - tubed pumps and patch pumps.

- Think of tubed pumps as little command devices. They come with inbuilt software, buttons, and a screen for pump control. These devices channel insulin through a length of tubing to an infusion set attached to your body, infusing it into your subcutaneous fat tissue via a cannula. Every 2 to 7 days, you need to replace the infusion set and the cannula to avoid infection and maintain control of glucose levels. The pump has to be worn 24/7, though it can be disconnected for up to 2 hours at a time for bathing, intimacy, or some types of exercise. It can be clipped to your clothing or placed beside you while sleeping.
- On the other hand, patch pumps are streamlined units consisting of only the insulin, the pump mechanism, and a battery. They are controlled by an external remote control or potentially a mobile phone. Affixed directly to your body, a patch pump needs to be replaced every three days just like most of the infusion sets of a tubed pump. They may offer more freedom in daily activities as there's no need to worry about the risk of tubing getting caught. However, you do need to carry a remote control or mobile phone all the time. And once you apply a patch pump, you have it on in that body site for 3 days, which may be more visible than a tubed pump, depending on what outfits you wear.

2.1 Tubed pumps

Popular tubed pumps include the MiniMed 780G, Tandem t:slim X2, Tandem Mobi, Kaleido, mylife Ypsopump, iLet Device, Dana pump and the twiist Pump.

- The MiniMed pump is the largest tubed pump, although size is usually not that big a deal if someone needs the features of a specific pump. The mylife YpsoPump and Dana pump are smaller tubed options. The Tandem Mobi, Kaleido, and twiist pumps are tubed systems that can also be worn on the body. The Tandem Mobi and Kaleido also offer shorter tubes, allowing the pump to be placed directly next to the infusion set.
- The majority of these tubed pumps have replaceable batteries that usually last for 1 to 4 weeks. On the other hand, the Tandem, Kaleido, and iLet pumps have a rechargeable battery, offering a more environmentally friendly option, but requiring regular charging.
- Insulin reservoirs are typically filled by hand from a vial of insulin, but the mylife YpsoPump and iLet Device can also be used with prefilled insulin

PumpCarts. These PumpCarts hold only 160 units of insulin, so they need replacing more frequently than reservoirs like the MiniMed pump, which can store 300 units.

- Tubed pumps usually have a screen and buttons to adjust insulin delivery. The Tandem t:slim X2 pump, mylife YpsoPump, and iLet device feature a touchscreen, while most other pumps rely on physical buttons for control. The Tandem Mobi, twiist pump, and Kaleido are app-controlled via smartphone, though both the Tandem Mobi and twiist pump also include a single button on the pump itself.
- These pumps can all be integrated with a sensor or algorithm via Bluetooth, which is why they're often used in automated insulin delivery systems today. These automated insulin delivery systems are discussed in separate modules of this Diabetes technology Expert Program.

2.2 Patch pumps

Patch pumps include the Omnipod DASH and Omnipod 5, TouchCare Nano pump, and the Equil pump.

Patch pumps are operated with a remote control or with an app on your phone.

- Always check whether your mobile phone is compatible and if the app is available in your country. For Omnipod 5 for example the app is only available in the US.

Most patch pumps are modular, with a reusable pump base that typically lasts up to 4 years, while the insulin reservoir needs to be replaced every 3 days. The Omnipod pumps are an exception, as they are fully discarded every 3 days.

The evolution in insulin pumps goes really fast. This content is updated in November 2025. For the latest updates, please visit the company's websites. Additionally, staying updated with resources like the Diabetotech blog can help you explore the evolving diabetes tech field.

2.3 Choosing an insulin pump

When choosing a pump, it's crucial to consider a few things.

- Check which insulin pumps are covered by your health insurance. Not all automated insulin delivery systems are reimbursed in all countries or even all regions or hospitals within the same country. Also, consult your diabetes

team on what technology they can offer that best suits your needs. Important considerations include the pump's ability to function as an automated insulin delivery system, your preference for a patch or tubed pump, size and weight of the pump, and needing to carry a remote control or mobile phone to control the pump.

- It's important to protect your pump and remote control appropriately. A silicone cover and screen protector are usually included and are recommended for use. A crack in your pump means it's no longer waterproof and needs to be replaced. Insulin pump systems typically have a warranty period of 4 years.
- Tubed pumps offer a variety of wearing options, and there's an array of helpful accessories such as customized pouches, belts, and specially-designed clothing and underwear. Many companies typically provide a belt clip or a holder at no extra cost. Remember to keep your pump within reach as you will need frequent access. Be sure to explore the range of unique and practical options available in various online specialty stores.
- When it comes to showering or bathing, most users choose to disconnect their tubed pump at the infusion set. For intimate activities, you have the option of either letting your pump hang if your catheter is long enough, or disconnecting it. Be mindful, however, that your pump should not be disconnected for more than two hours, as this could leave you without insulin, increasing the risk of ketoacidosis.
- Lastly, bear in mind that the main issue with insulin pumps is making sure the insulin is getting into the body safely. Most infusion sets need to be changed every 3 days to reduce the risk of an infection and keep glucose levels in control. Going without insulin because it's not being absorbed from the infusion site can increase the risk of diabetic ketoacidosis, which is a serious health emergency.

Now you have a general feeling on how insulin pumps work, which insulin pumps are available and what the differences are. This may seem like a lot of information, but once you really get to grips with it, you will see that each insulin pump works largely the same and is reasonably intuitive. Most people on an insulin pump will be perfectly capable of operating their pump themselves, after all, they work with it 24/7. So don't be worried about starting insulin pump therapy. Think of it like using a mobile phone, once you get the hang of it, the functionality becomes second nature.

3. Insulin settings

Welcome to this comprehensive video about insulin pump settings!

An insulin pump offers an intricate way to deliver insulin. It allows you to control both your basal and bolus insulin levels with only short acting insulin.

In this discussion, we'll explore the various settings and how to set basal and bolus insulin settings as you switch from insulin pens to a pump.

It's important to note that most guidelines about insulin pump settings are opinion-based, and that these videos are not intended as medical advice. Always consult with your healthcare provider before changing your treatment.

3.1 Overview of insulin settings

In an insulin pump, you can adjust basal insulin delivery for each half-hour or hour-long segment throughout your day.

- In most pumps, you can set up multiple different basic profiles to select from.
- There's also a feature to choose a temporary basal rate, which allows you to adjust your insulin release for a specific period of time.
- You can also stop or suspend your insulin pump at any time. However, be cautious about stopping the pump for longer than two hours, as it could lead to ketoacidosis.

As for bolus insulin, the pump gives two options:

- a manual bolus, which lets you determine the insulin amount you want to administer,
- and a bolus from the bolus calculator, which computes the necessary insulin based on your carbohydrate intake, blood sugar level, carbohydrate ratio, insulin sensitivity, target value, and active insulin.

Furthermore, you can choose among various types of boluses like a simple bolus, an extended bolus, or a combined bolus, that could be particularly useful for meals high in fat.

3.2 Basal insulin rate

Most individuals with type 1 diabetes using insulin pens follow a basal-bolus regimen. This consists of one or two injections of long-acting insulin and three or more injections of rapid-acting insulin for mealtime carbohydrate intake.

In transitioning to an insulin pump, we generally calculate the total daily insulin dose, by making the sum of both long and short-acting insulin, and set the pump to deliver 80% of this total daily insulin dose. This reduced dose is because the insulin seems to work more efficiently via the pump, and we want to avoid hypoglycemia.

As an example, let's consider Anne, who has type 1 diabetes. Every day, she injects 45 units of Lantus before bed and on average, she has a total of 30 units of Novorapid with her meals. That's 75 units in total. If she switches to an insulin pump, she'll need to reduce her total daily insulin dose to 60 units.

This insulin amount is then divided into doses of basal and bolus insulin to be given by the insulin pump. For most people, the starting ratio of basal to bolus is about 50/50%, but it can vary depending on individual dietary habits. This means that the basal daily insulin dose will be half of the previously calculated total daily insulin dose.

Now that you know the total basal insulin dose, you can divide that by 24 and set that as a fixed hourly rate. In our previous example, we determined that Ann's total daily insulin requirement would be 60 units when switching to an insulin pump. From this, her total basal insulin need can be calculated to be 30 units per day. By dividing this amount by 24 hours, we arrive at a calculated basal insulin rate of 1.25 units per hour. Most healthcare providers are accustomed to further fine-tuning the basal rate to align with the circadian insulin needs of their patients.

- For manual insulin pumps, it is common practice to set a lower basal insulin rate at night, for example 20% lower, and a higher basal rate during the day to better match the body's varying insulin requirements.
- For people with known dawn phenomenon, a higher basal rate can be set between, say, 2AM and 7AM.

To evaluate basal insulin, you can monitor blood sugar levels. This can be done either with finger sticks or a glucose sensor, but using a glucose sensor makes the process easier.

- At night, this is relatively straightforward as no additional factors like eating or exercising affect your blood sugar.
- During the day, it's a little more complex. One way is to occasionally skip or delay a meal for a few hours and see if there's a persistent change in blood sugar levels. A rise or fall of more than 30 mg/dl (or 1.7 mmol/l), may indicate a need for basal insulin adjustment.

Once the basal rates are set, the bolus calculator parameters can also be defined.

3.3 Bolus calculator parameters

Bolus calculator parameters include the carbohydrate ratio, insulin sensitivity factor (also called correction factor), bolus calculator target value, and duration of insulin action. We'll discuss them one by one.

Insulin-to-carbohydrate ratio

The Insulin-to-Carbohydrate ratio or carb ratio is the quantity of carbohydrates that can be covered by one unit of insulin. To determine the amount of insulin to bolus for a set amount of carbs, patients need to count their carbohydrate intake during meals and then divide it by the calculated carb ratio.

For example if you're going to eat 50 grams of carbs and your carb ratio is 5 grams per unit, you will need to bolus 10 units of insulin.

The ratio can be established in a few ways:

- by employing the 450-rule,
- through a dietician who calculates the carbohydrate intake per meal and then divides it by the amount of insulin units injected,
- or by taking a certain percentage of the total daily insulin dose.

The 450-rule means that you can divide 450 by the total daily insulin dose to calculate the carb ratio.

For example, if Ann's total daily insulin dose was 60 units, and using the 450-rule, her carbohydrate ratio would be 7.5 grams per unit. Hence, for a meal of 15 grams, she would need 2 units of insulin.

Alternatively, some healthcare providers use the 300 to 500-rule to calculate the carb ratio.

Another method involves a dietician calculating the actual carbohydrate intake, which is then divided by the units of insulin injected.

Lastly, one can take a certain percentage of the total daily dose for meal boluses. In the case of fixed portion meals, it is typically recommended to use 50% of the total daily insulin dose for all meals. Given that more insulin is usually required during breakfast, it can be divided as follows: 20% for breakfast, 15% for lunch, and 15% for dinner.

To evaluate the carbohydrate ratio, check the blood sugar level 3-5 hours after an insulin bolus. It should return to pre-meal levels. If it is consistently higher or lower, consider adjusting the carbohydrate ratio by 10%.

Insulin sensitivity

The second parameter for the bolus calculator is insulin sensitivity. This reflects the amount the blood sugar level drops per unit of insulin, measured in mg/dl or mmol/l. The 1800-rule (or 100-rule in mmol/l) is commonly used, but a more aggressive setting could be the 1500-rule (or 90-rule), and a less aggressive setting might be the 2000-rule (or 120 to even 130-rule).

Take Ann's case for example: with a total daily insulin dose of 60 units, her insulin sensitivity can be calculated using the 1800-rule. 1800 divided by 60 equals 30. Therefore, setting the insulin sensitivity to 30 mg/dl per unit means that a unit of insulin will decrease the blood sugar level by 30 mg/dl.

Glucose target

The bolus calculator also requires a glucose target, which could be a fixed number or a target range.

Depending on the type of insulin pump, the bolus calculator may use the target value differently.

- Most pumps will correct to the middle of the target range.
- Some pumps, like the Omnipod DASH, also have a 'Correct Above' value, indicating when a correction will be applied.

- Some pumps also offer an option for reverse correction, subtracting insulin if the current blood sugar level is lower than the target.

It is generally recommended to set a fixed target value or a target range of 90-100 mg/dl (5-5.5 mmol/l). However, this should be individualized.

Duration of insulin action

Lastly, you need to set the duration of insulin action. This parameter determines how long the insulin stays active, and thus how much of the previous insulin bolus is still in effect. To avoid insulin stacking, it's crucial for the pump to take this into account. Typically, this is set to 3 to 5 hours.

So now you understand how to set the basal insulin and the parameters for the bolus calculator.

3.4 Do's & don'ts

- On the first day of using an insulin pump, basal insulin from the insulin pens may still be present. It is recommended to inject only half the long-acting insulin the day before starting the pump. For users of Toujeo or Tresiba, it may be best to switch to Lantus a week in advance due to its shorter action profile. If unsure, it's safer to reduce the basal insulin rate when starting the insulin pump to prevent hypoglycemia.
- When starting on an insulin pump, it's usually necessary to set a limit for insulin delivery, i.e., the maximum basal and maximum bolus. Typically, it's recommended that the maximum basal insulin delivery should be double the highest set basal insulin delivery, and the maximum bolus should be the largest bolus expected for a large meal.
- All insulin pumps allow users to set various profiles for different situations. The number of basal profiles that can be set varies depending on the pump. Some pumps only store the basal insulin in a profile, while others, like the Tandem Pumps, also allow adjustment of the carbohydrate ratio and insulin sensitivity per profile.
- There are also several strategies for adjusting your insulin needs with temporary basal rates.
 - These should be evaluated individually. In some cases you could consider a temporary higher basal rate, for instance:
 - a 50% higher basal rate for 8 hours after eating a high-fat meal

- an 80% higher basal rate for when you're ill
- a 50% higher basal rate 3-6 hours after receiving a corticosteroid injection
- an 80% higher basal rate from 1-2 hours before a scheduled stressful event
- a 40% higher basal rate during prolonged inactivity, like a long car ride
- a 50% higher basal rate in the week before your menstruation, some people also set a 30% lower basal rate during their menstruation
- In other cases you can consider a lower temporary basal rate, for instance:
 - a 50% lower basal rate from 1-2 hours before extensive activity like housework or yard work
 - a 30% lower basal rate for 6-10 hours after exercise to prevent late onset hypoglycemia
 - or a 40% lower basal rate after consuming alcohol for 2 hours per drink

Lastly, remember to constantly monitor your blood sugar level. The insulin pump's primary advantage is the precision and customization it offers. To make the most of it, you should adjust settings based on the observed patterns. You may also need to tweak these settings frequently, at least in the beginning, to achieve optimal glucose control.

Remember that every person is different, and while these guidelines may apply to many, there's no 'one-size-fits-all' solution in diabetes management. It requires personal fine-tuning and learning by trial and error under the supervision of your healthcare provider.

Keep in mind that while insulin pumps can make management easier and more precise, they also require a deeper understanding and commitment to managing your diabetes. In some ways, it can be more demanding because of the necessary knowledge and continual adjustments.

We hope this presentation has provided you with some valuable insights about transitioning from insulin pens to an insulin pump. We encourage you to reach out to your healthcare team for further support and guidance during this transition.

4. Reservoir and infusion set replacement

Welcome to this lesson about reservoir and infusion set replacement in insulin pumps.

This tutorial will guide you through the available infusion sets, essential steps of replacing your insulin reservoir and infusion set properly, and adverse infusion site reactions.

4.1 Different infusion sets

While patch pumps usually work with 1 fixed cannula, tubed pumps offer a wider selection of infusion sets and tube lengths:

- **Type of Cannula:** You can choose between angled and straight plastic cannulas, or a straight steel cannula, each coming in various lengths. The standard choice is a straight plastic cannula that can remain in place for three days. However, for individuals frequently experiencing discomfort from a folded subcutaneous cannula or who have allergies, a steel cannula might be a more suitable alternative. Remember, steel cannulas generally need replacement every two days instead of three.
- **Length and Angle of Cannula:** Usually, the shortest cannula is selected, but if you notice insulin leakage at the infusion site, a longer or angled cannula might be a better solution. An angled cannula could be particularly beneficial for those with less subcutaneous fat or those who find straight cannulas uncomfortable.
- **Tube Length:** Tubed pumps allow the option to select tubes of different lengths, adding another layer of customization to your insulin delivery system.
- MiniMed has infusion sets that offer extended wear, lasting up to 7 days.
- Ypsopump's mylife orbit infusion sets come with a unique feature - the tubing can rotate 360-degrees around the skin-colored adhesive patch.

An overview of all available infusion sets for tubed pumps can be found in the download section below this video. Remember, the goal is to choose an infusion set that ensures comfort, minimizes complications, and optimizes insulin delivery. Your healthcare provider can help guide your choice based on your lifestyle, physical characteristics, and personal preferences.

4.2 Supplies

Before starting, make sure you have the following items handy:

- Insulin: depending on the pump you use, you can use a pre-filled insulin PumpCart, or draw insulin from an insulin Penfill or Vial using a needle into the pump reservoir.
- Infusion set and tube: These might be bundled together or separate.
- An insertion system may be used to insert the infusion set into your skin
- And alcohol wipes or skin prep wipes for sanitization.

4.3 Four-step procedure

The process of changing an insulin reservoir and infusion set in a tube pump generally involves four stages:

- Refilling or swapping the insulin reservoir
- Filling the tubing
- Placing the infusion set
- And filling the cannula under the skin

If you're using a patch pump, there's no tubing to fill. You will have to fill the patch pump with insulin, follow the priming process, place the patch pump on your skin and insert the cannula.

After replacing your insulin and infusion set, it's crucial to check your blood sugar level about 2 hours later or keep an eye on your glucose sensor for the next few hours to make sure the change was successful. It's best practice to avoid changing your infusion set before bedtime unless you're prepared to wake up two hours later to check your blood sugar level.

4.4 Infusion site management

Consider these guidelines when selecting and preparing an infusion site:

- Choose an Ideal Spot:

- Opt for the abdomen, back of the upper arms, or lower back. Avoid the upper legs due to unpredictable absorption rates, particularly during exercise.
 - On a limb, mount the patch pump vertically. The pump's window can point either direction. For the back, abdomen, or buttocks, position it horizontally or slightly angled.
 - Maintain a minimum distance of 5 cm or 2 inch from the navel and the last infusion site. Avoid placing under belts, near bones, or on scars, injuries, and skin conditions.
 - Rotate your injection sites regularly to prevent lipodystrophy, a condition that affects the fat under the skin. Creating a simple rotation schedule will help ensure consistency and reduce the risk of developing this problem.
- Preparation Steps:
 - Clean the skin with an alcohol wipe to ensure adherence. Allow it to air dry and ensure no soap or lotion residue or excess body hair.
 - Avoid applying sunscreen or insect repellent near the site as it could affect the pump's function.
 - For sensitive skin or allergies, opt for barrier wipes like Skin Prep or Conveen wipes, instead of alcohol wipes.
 - Consider additional adhesive like Skintac if the infusion set tends to come loose.
 - Slight discomfort upon insertion may occur. Ice or anesthetic cream may help, but ongoing discomfort could indicate the cannula touching a nerve or muscle, necessitating removal and reinsertion.
- Routine Checkups:
 - Inspect your infusion site twice daily for signs of irritation or infection such as pain, swelling, redness, warmth or insulin leakage. In these cases, reposition the infusion set: "When in doubt, change it out".

4.5 Do's & don'ts

Here are some key points to ensure efficient insulin delivery and prevent issues:

- The main risk associated with insulin pumps is ketoacidosis. Those new to using an insulin pump should be educated that if they experience prolonged hyperglycemia (for example more than 2 hours above 270 mg/dl or 15

mmol/l) without a clear reason, they should always consider a problem with the infusion set and not just sleep through it. The mantra is “if in doubt, change it out”.

- Managing Air Bubbles:
 - Let insulin reach room temperature before filling the reservoir to minimize air bubbles.
 - Ensure you remove any large air bubbles during the filling process. Small bubbles won't harm, but an air bubble over 2.5 cm length equals a 0.5 unit insulin gap, so it's best to eliminate them.
- Timely Infusion Set Replacement:
 - Replace the plastic cannula every 3 days and steel cannula every 2 days. If you're using an Extended Wear Infusion Set, replace it every 7 days.
 - Some pumps won't halt insulin delivery after 3 days, so set an alarm to avoid missing the replacement, or purposely put just the amount of insulin you expect to use for 3 days, plus the amount to fill the tubing, into the pump, to ensure you will change the set on time. Other pumps, like the Omnipod DASH and Omnipod 5, automatically stop after a set time.
 - Carefully remove infusion sets or pods to avoid skin irritation, and clean any leftover adhesive with soap and water or a specialized product.
- Choosing the Right Filling Volume:
 - Each type of tubing and cannula requires a specific filling volume as per its user guide.
 - Be aware that short steel cannulas might not need filling.
 - Do not overlook filling the cannula; otherwise, you may miss the initial 0.5 unit of insulin.
 - You can check the necessary filling volume of the cannula by looking at the manual of the infusion set or at the overview in the download section.
- In the rare event of the infusion set coming loose from your body, you can use an overpatch to secure it more firmly. It is recommended to leave an opening in the overpatch to allow for disconnecting the infusion set when needed, such as during showering.

Remember, it's not uncommon to encounter issues during infusion set insertion, like a kinked cannula or premature insertion mechanism, but it's usually a minor setback. You can always restart with a new set. If you see blood post-insertion, reinsert the set. If you're out of practice, instructional videos on YouTube can be helpful. Good luck!

5. Pump alarms

Welcome to this lesson on insulin pump alarms.

We'll discuss alarm types and some general do's & don'ts.

Although these alarms vary across different pumps, they share certain common characteristics.

5.1 Alarm types

Insulin pumps generally categorize alarms into two priorities:

- High-priority alarms: These include an empty insulin reservoir, depleted battery, technical faults, etc., leading to a halt in insulin delivery.
- And Lower-priority alerts: These are typically related to a low insulin reservoir, low battery, blockages, etc. The pump continues insulin delivery but requests problem resolution.

Each alarm requires acknowledgment and action. If ignored, alarms will progressively get louder. You can often customize alarm volume and vibration settings and can temporarily silence lower-priority alarms. The pumps usually allow setting up one or more reminders.

5.2 Do's & don'ts

- Familiarize yourself with potential alarms to avoid panic when one occurs. Instruction manuals outline all alarms and the appropriate actions, so consider saving a digital copy on your phone.
- Contact the pump manufacturer's customer service department if needed, keeping in mind that not all offer 24/7 support. It's worth creating a Contact in your mobile phone for your insulin pump company, and including their hours of operation as a note on the contact.
- Some alarms, like the "screaming pod" from Omnipod, can be tricky to deactivate without the remote control. In such cases, use a paperclip to disable the chip card through the pod's back opening.
- If you're temporarily not using your pump, switch it to storage mode to prevent constant alarms. Each pump has its unique procedure for this, outlined in its operating instructions.
- It's crucial to be aware that in the event of a pump malfunction, insulin delivery remains essential. To ensure preparedness, it is recommended to have your rapid- and long-acting insulin pens easily accessible as a backup. Create an emergency plan outlining the required insulin doses in case of pump failure, and make sure to update this plan regularly based on your

current insulin needs. Being prepared for unexpected situations is vital, as pump problems can arise at any time.

- Additionally, it's wise to keep a record of all your basal rates and bolus ratios in a note on your phone, which is synced to the Cloud, or email yourself a copy whenever you make any changes. This way, if you need to set up a new pump, you will have a record of your settings readily available. Taking these precautions ensures you are well-prepared to handle any potential issues with your insulin pump effectively.

Now, you're equipped with some essential knowledge about pump alarms and how to respond. If an alarm triggers, stay calm and check the screen instructions or refer to the user manual or customer service. Good luck!

6. Case story

Welcome to our final chapter, where we have a special video for you to watch. In this segment, we hear from Andrea, an insulin pump user with over 19 years of experience managing type 1 diabetes. Andrea shares her personal transition story from using insulin pens to embracing an insulin pump, providing valuable insights into this significant change.

In the video, Andrea discusses her decision to switch to an insulin pump due to frequent low blood sugars, her initial emotional and overwhelming impressions of the pump, and the challenges she faced in the beginning, such as dealing with a kinked cannula after just two hours of training. She emphasizes that adapting to the pump can be a substantial adjustment, and feeling overwhelmed is entirely normal. Despite the initial difficulties, Andrea speaks highly of the pump, highlighting how it has significantly improved her life.

To watch Andrea's inspiring journey, click on the link below this video.

<https://www.youtube.com/watch?v=wVr89eDEKqc> – 17 min

Congratulations on completing this comprehensive introduction to insulin pump therapy! Your dedication to understanding this essential aspect of diabetes management is commendable and will undoubtedly empower you to support individuals navigating their diabetes journey.

By acquiring vital knowledge, you have equipped yourself to guide and assist those living with diabetes effectively. This is truly an outstanding achievement!

Take a moment to reflect on what you've learned. When you're ready, consider exploring more about specific insulin pumps through our additional modules.

Thank you for joining us on this enlightening journey. Your commitment is making a positive difference in transforming diabetes from a life-limiting condition to a manageable aspect of daily life. Great job!